

Original papers

3D skeletonization and phenotyping for soybean root system architecture using a bio-inspired algorithm

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ABSTRACT

Characterizing root system architecture (RSA) is essential for understanding plant acclimatization and guiding breeding strategies to enhance stress tolerance and optimize resource uptake. Although 3D root analysis provides significantly more detailed and structurally informative insights than conventional 2D methods, the development of robust and quantitative tools for 3D root phenotyping has been hindered by challenges such as data complexity, noise, and root overlap. In this study, we present a biologically inspired skeletonization framework that segments root architectures by tracing root growth trajectories. The primary objective is to enable anatomically accurate extraction of RSA traits from 3D point clouds. Our method begins by segmenting the primary root through shortest-path extraction and tangent-plane-based clustering. Lateral root initiation points are then detected, and candidate paths are grown using a bionic pathfinding strategy with adaptive parameters; an optimal, non-overlapping skeleton is selected through clustering and combination sorting, and finally refined via an inward back-tracing procedure to improve junction connectivity. To support downstream phenotyping, we compute root length and angle from the segmented skeletons, and reconstruct anatomically faithful tubular meshes for each lateral root to analytically estimate surface area and volume. Our method achieved high accuracy across multiple traits, including an F_1 score of 0.88 for lateral root numeration, R^2 values of 0.992 and 0.987 for primary and lateral root length estimation, respectively, and strong agreement in surface area ($R^2 = 0.953$) and volume ($R^2 = 0.912$) validation against reference methods. Overall, our method offers a robust and biologically meaningful solution for 3D root phenotyping. The extracted traits provide plant breeders with critical insights for genotype selection and offer plant scientists a powerful tool to evaluate the effects of agronomic treatments and environmental interventions.

1. Introduction

Root system architecture (RSA) plays a fundamental role in plant growth, resource acquisition, and environmental acclimation. A well-structured root system enhances water and nutrient uptake, which is crucial for mitigating abiotic stresses such as drought and nutrient deficiency. Specific RSA phenotypes, such as increased root depth, greater lateral root density, and narrower branching angles, have been associated with improved resource acquisition under stress conditions (Bucksch et al., 2017). Consequently, phenotyping RSA is essential for breeding programs aimed at developing stress-tolerant and high-yielding crops (Jiang et al., 2024). However, traditional RSA

phenotyping has largely relied on 2D imaging, which fails to capture the full complexity of root structures, limiting its applicability for in-depth morphological analysis (Chandnani et al., 2023; Gaggion et al., 2021; Yasrab et al., 2019; Xu et al., 2022). With advancements in 3D imaging technologies, researchers can now reconstruct root systems in the 3D space, allowing for more accurate and biologically meaningful trait extraction (Liu et al., 2021a; Morris et al., 2017).

A variety of methods have been developed for acquiring 3D root system data, each customized for different experimental setups and plant species. X-ray computed tomography (CT) enables non-destructive imaging of roots within the growth medium, preserving their internal

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structure without physical disturbance (Ju et al., 2024; Rogers et al., 2016). However, its limited field of view constrains its applicability, making it most suitable for studying small potted root systems or plants in their early growth stages (Mairhofer et al., 2015). Magnetic resonance imaging (MRI) offers similar advantages but is primarily limited to water-rich environments (Pflugfelder et al., 2021). Laser scanning and structured light imaging have been widely used for capturing excavated roots, allowing for rapid point cloud generation with high spatial accuracy (Montagnoli et al., 2024). More recently, multi-view stereo (MVS) and photogrammetry techniques have gained popularity, leveraging standard RGB cameras to reconstruct 3D root models with minimal hardware requirements (Liu et al., 2021a; Wu et al., 2023). Despite these advances, the challenge of extracting meaningful traits from 3D root system data remains an open problem.

Recent advances in plant root system phenotyping are still primarily focused on 2D analysis, largely driven by the booming development of deep learning-based 2D image recognition methods. For instance, RootEx, a recently published method, introduces an automated pipeline for extracting barley root systems from 2D high-resolution images, enabling high-throughput trait computation (Dadi et al., 2025). Similarly, ChronoRoot 2.0 presents a deep learning framework that temporally tracks root and shoot development in 2D time-lapse images, offering dynamic phenotyping capabilities (Gaggion et al., 2025). EU-GAN leverages generative adversarial networks to inpaint occluded or missing root segments in 2D scans, enhancing the completeness and quality of phenotypic data (Xie et al., 2025). Other recent contributions, such as landmark grouping-based root pose estimation (Berrigan et al., 2024) and enhanced segmentation techniques for in situ crop roots (Banet et al., 2024), further exemplify the continued emphasis on 2D RSA phenotyping.

In contrast, 3D RSA phenotyping has progressed more slowly due to the inherent challenges of acquiring, processing, and analyzing volumetric data. Despite its slower momentum, 3D analysis offers a more comprehensive characterization of root morphology, enabling the extraction of spatial traits that are inaccessible in 2D. Among various techniques, skeletonization plays a central role by transforming complex root structures into simplified topological graphs that preserve key features such as root length, branching angles, and depth. However, 3D skeletonization of plant structures remains a technically challenging and relatively underexplored area—especially for dense and intricate root systems. Given the limited number of studies specifically addressing 3D RSA, we also review skeletonization techniques developed for above-ground plant structures such as shoots, which face similar geometric and topological complexities. Laplacian-based mesh contraction, for example, has been widely applied to reconstruct shoot architectures like those of cotton and maize, simplifying point clouds by iteratively minimizing curvature (Sun et al., 2021; Liu et al., 2021b). Reeb graph-based methods, such as SkelTre, extract topological graphs from point cloud geometry and are robust to noise and incomplete data (Bucksch et al., 2010). L1-median skeletonization, though effective at preserving global topology, requires intensive optimization and can be computationally expensive for dense RSA datasets (Feng et al., 2024). These longstanding efforts underscore the complexity of 3D plant analysis and the continued need for biologically informed, scalable skeletonization approaches.

Trait computation of surface area and volume for individual roots primarily depends on accurate RSA segmentation. Compared to skeletonization, plant segmentation has gained more attention in recent years with the rise of deep learning-based 3D methods, extending beyond root systems to encompass a wide range of above- and below-ground structures. Segmentation was once highly challenging, especially before the adoption of deep learning for 3D point cloud analysis. However, training such models typically requires large, well-annotated datasets, which remain difficult to obtain in plant phenotyping. To address this, early work employed traditional non-deep learning techniques. For example, Sun et al. used region-based classification and

density-based clustering on LiDAR point clouds to extract boll count, volume, and spatial distribution in cotton (Sun et al., 2020), while Jiang et al. later adopted PointNet++ with Euclidean clustering for finer segmentation of cotton bolls and branches (Qi et al., 2017; Jiang et al., 2025b). Li et al. proposed downsampling strategies that preserved plant contours while reducing computational load (Li et al., 2022), and Luo et al. introduced Eff-3DPSeg, a semi-supervised, annotation-efficient framework for shoot segmentation (Luo et al., 2023). In root systems, Zhou et al. showed that effective taproot segmentation can be achieved with minimal annotation in simple architectures (Zhou et al., 2024). More recently, Jiang et al. proposed an organ-level segmentation pipeline for apple trees using PointNeXt and SoftGroup++, combined with graph-based postprocessing, achieving high accuracy with limited training data (Qian et al., 2022; Jiang et al., 2025a).

In this work, we introduce a bionic root system parser, a bottom-up skeletonization framework inspired by natural root growth patterns. Our method dynamically constructs the skeleton by following the intrinsic growth trajectory of lateral roots, ensuring accurate representation while preserving the root system's hierarchical structure. Our main contributions are as follows: (1) *Bionic pathfinding algorithm for noise-robust and artifact-free skeletonization*: By assuming lateral roots grow outward with a smooth trajectory, we restrict the pathfinding process to a localized retrospective search space, effectively reducing noise interference and preventing the inclusion of spurious paths. (2) *Adaptive pathfinding to avoid breakages*: Our algorithm dynamically adjusts search parameters, including step size and angular range. This prevents discontinuities in sparsely sampled regions (e.g., lateral root tips) and avoids excessive contractions in dense regions (e.g., branching junctions). (3) *Combination sort for overlap-aware path selection*: To resolve ambiguous cases where multiple paths overlap or converge, we introduce a global overlapping list that iteratively selects the best lateral root path combinations. This ensures a non-redundant and topologically accurate skeleton representation.

Through validation on 3D root system datasets, we demonstrate that our bionic root parser outperforms conventional skeletonization methods, producing high-precision, phenotypically meaningful skeleton representations. By improving root segmentation and skeletonization accuracy, this work contributes to the broader goal of automated 3D RSA phenotyping, facilitating more robust genotype-to-phenotype associations in plant science and breeding applications.

2. Materials and methods

2.1. Dataset

The soybean plants were grown at the Macdonald Campus Farm of McGill University (45.4329° N, 73.9326° W), in field soil without any added fertilizers or chemical treatments during the growing period. A total of 24 plants were sown on July 5, 2024. Root systems were extracted using shovelomics (Burrige et al., 2016), with 12 plants harvested on July 20 (15 days after sowing) and the remaining 12 plants on August 5 (30 days after sowing). The dataset includes two soybean genotypes – Bayfield and Madoc – developed by the Ontario Agricultural College at the University of Guelph. These cultivars represent distinct genetic backgrounds widely used in Canadian soybean breeding programs.

Our dataset comprises 24 3D RSA point clouds, eight of which are shown in Fig. 1. The data were acquired using a turntable-based photogrammetry system equipped with two Sony Alpha 6400 mirrorless cameras, each fitted with a Sony E PZ 16–50 mm f/3.5–5.6 OSS lens, to capture stereo images of root systems inserted into pots in an inverted orientation. A triggerbox automated the camera shutters at one-second intervals to synchronize image capture. For each root system, a total of 100 stereo images were taken per camera. A background removal algorithm was applied prior to 3D reconstruction with Agisoft Metashape (v2.2.1). Finally, coded markers of known dimensions were included for real-world size calibration.

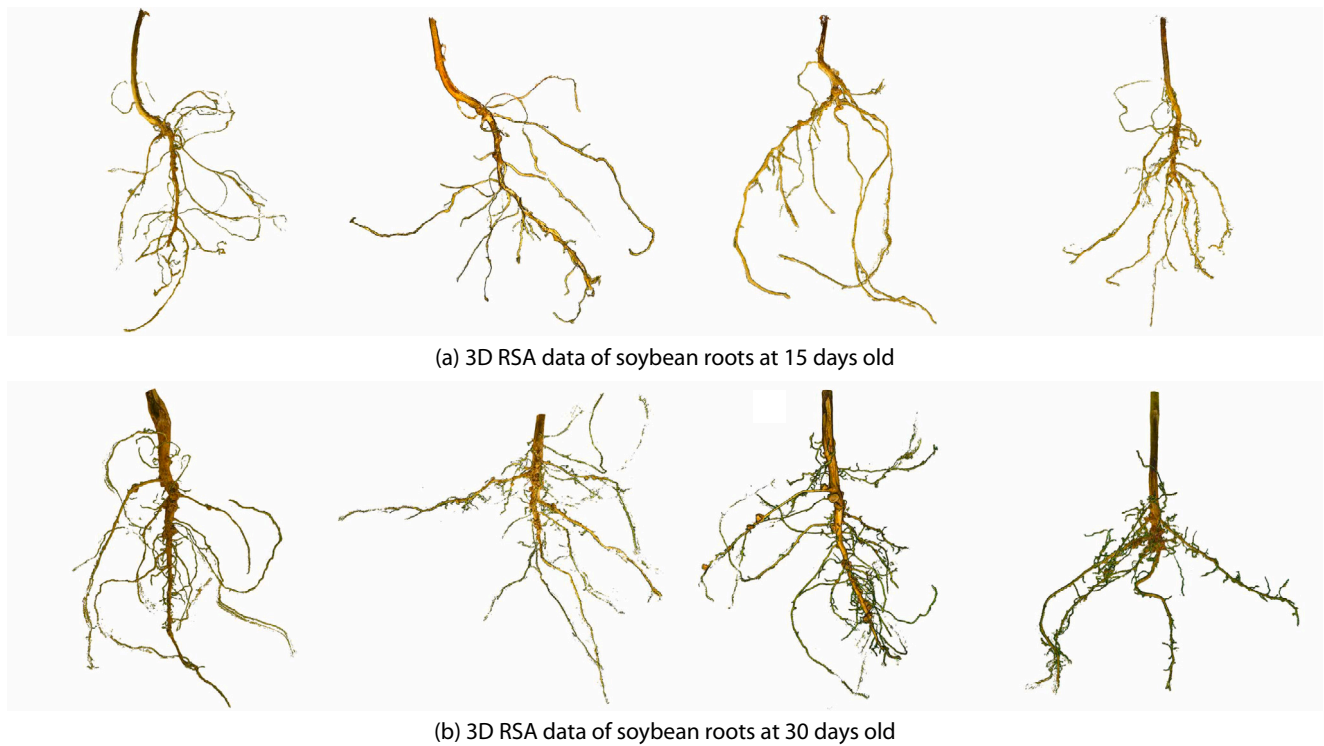


Fig. 1. 3D RSA Dataset. This dataset includes 24 3D soybean RSA point clouds. (a) Twelve samples were collected at 15 days post-germination, with four representative examples displayed. (b) Another twelve samples were collected at 30 days post-germination, with four examples shown.

2.2. Overview

Our data processing pipeline is illustrated conceptually in Fig. 2. The input consists of 3D RSA data reconstructed using Structure from Motion (SfM) (Schonberger and Frahm, 2016), as shown in Fig. 2a. To ensure compatibility across diverse root system architectures, each point cloud is first normalized to fit within a unit bounding box and translated to be centered at the origin (0, 0, 0), as illustrated in Fig. 2b. A KDTree structure (Bentley, 1975) is employed for nearest-neighbor searches and to compute the average nearest-neighbor distance, denoted $\bar{d}$. Specifically, for each point in the root system, the distance to its closest neighbor is calculated, and $\bar{d}$ is defined as the mean of these distances. This adaptive scaling parameter replaces fixed distance thresholds, ensuring that all subsequent geometric operations, such as search radii and filtering, are automatically tuned to the underlying point density of the 3D RSA dataset. This promotes robust generalization across root systems with varying scales and sampling resolutions.

Fig. 3 provides an overview of our root parsing pipeline, the main contribution of this work, which bridges the normalization step shown in Fig. 2b with the final skeleton and segmentation results shown in Fig. 2c. This algorithm retrospectively traces and skeletonizes the root system morphology while preserving structural integrity. The process begins with preprocessing the 3D RSA data, followed by primary root segmentation (Section 2.3). Removing the primary root point cloud isolates the lateral roots, allowing them to be geometrically separated. Next, we identify junctions between the primary and lateral roots, establishing a list of starting points for lateral roots (Section 2.4).

From each starting point, the algorithm dynamically grows multiple outward path candidates, ensuring that there is one candidate that could best represent each corresponding lateral root (Section 2.5). To efficiently determine the final path combination across all lateral roots, we construct a global overlapping list, which groups starting points that may generate overlapping lateral root paths. These overlapping groups are then visualized in distinct colors, as illustrated in Fig. 3 (Build

global overlapping list). The algorithm then iterates through the local overlapping points within the global list, optimizing the selection of lateral root paths for each local subset (Section 2.6). By stacking these locally optimized path combinations, we reconstruct a high-precision, segmented lateral root skeleton.

Finally, we apply a back-tracing algorithm to refine the skeleton, recovering missing regions that were initially excluded for better lateral root identification (Section 2.7). This final step ensures a more accurate and biologically meaningful representation of the root system. By associating surrounding points with the extracted skeletons, we further obtain a point cloud-wise segmentation of the first-order lateral roots. In addition, to facilitate clarity and reproducibility of our mathematically and computationally driven framework, we provide a comprehensive summary of all mathematical symbols used throughout this work in Table 1.

2.3. Primary root identification

Our primary root segmentation and skeletonization procedure is illustrated in Fig. 4. We begin with normalized 3D RSA point clouds as input. Two endpoints on the primary root are manually selected (Fig. 4a), and a Dijkstra's shortest-path algorithm (Cormen et al., 2022) is applied to compute the minimal geodesic distance between the two points along the root surface (Fig. 4b). Next, a local tangent plane is defined at each point along the computed path (Fig. 4c). Tangent vectors are obtained by differentiating the path coordinates, applying Gaussian smoothing, and normalizing them to unit length (Algorithm 1). This smoothing reduces noise and ensures gradual directional transitions, providing stable reference planes throughout the path.

We then project neighboring points onto each tangent plane and apply Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) to separate the primary root from lateral roots (Malzer and Baum, 2020). HDBSCAN identifies stable clusters by iteratively assessing cluster persistence across a range of density thresholds. Neighboring points are defined as those within a radius of

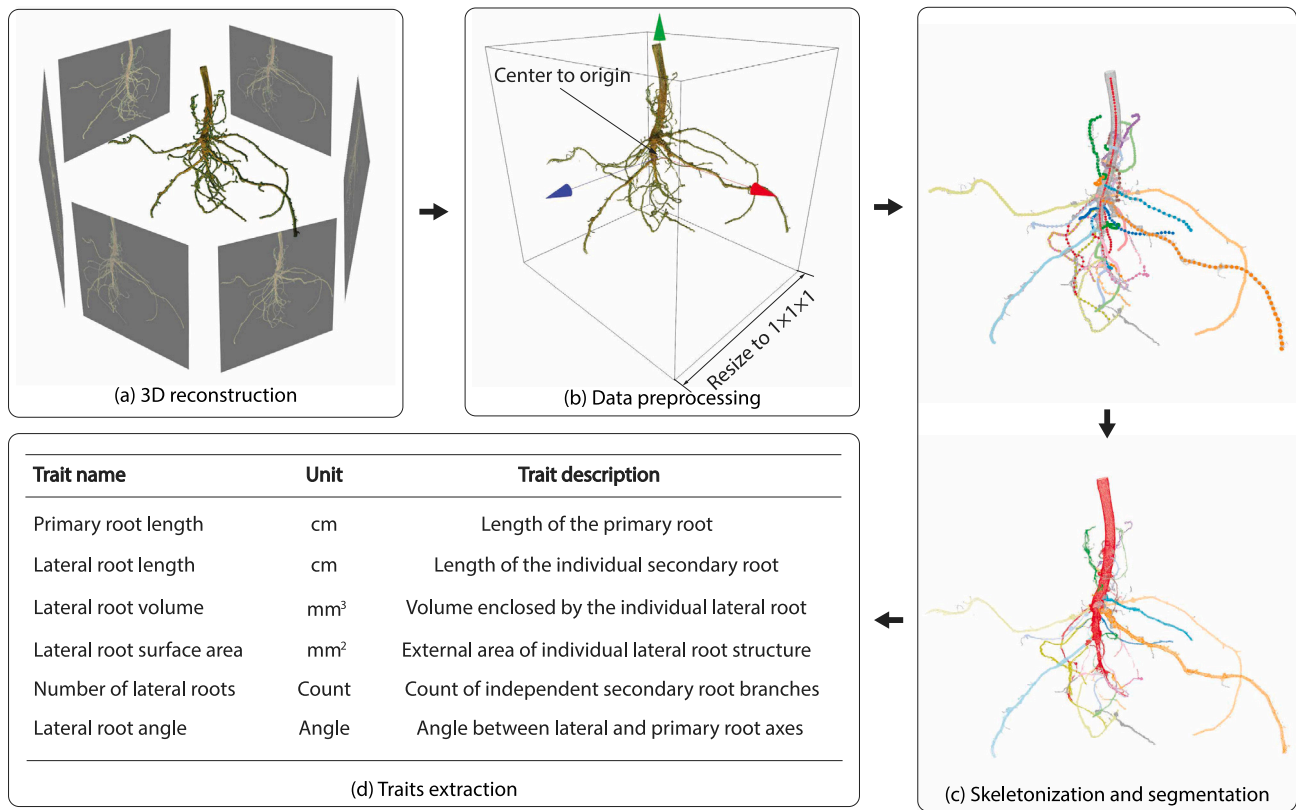


Fig. 2. Method overview. (a) The 3D RSA data is reconstructed from multiview stereo imaging. (b) The reconstructed data undergoes preprocessing steps, including normalization to fit within a unit cube and centering at the origin. (c) The normalized RSA is then passed through our parsing pipeline that extracts a geometric skeleton and segments individual root branches. (d) Key traits – such as lateral root counts, root lengths, volumetric measures, and angular distributions – are computed for quantitative analysis.

Table 1
Summary of mathematical symbols used in the bionic RSA skeletonization and segmentation framework.

Group	Symbol	Description
Distance and path primitives	$\bar{d}$	Average nearest-neighbor distance of the point cloud, used as a dynamic scaling factor.
	p	A single lateral root path (skeletonized polyline) traced from a given starting point.
	P	Total set of path candidates generated from a single starting point.
	P'	Reduced set of candidate paths after similarity-based clustering for a single starting point.
	$\mathcal{P}$	Complete set of path candidates across all starting points.
Point cloud associations	s	Set of surrounding points (point cloud subset) associated with a given path p .
	S	All surrounding point sets corresponding to paths in P .
Clustering and indexing	K	Number of clusters in similarity-based clustering (e.g., $K = 4$).
	N	Total number of lateral root starting points in the root system.
	N_i	Number of branches (starting points) in the i th local overlapping group.
	l	Total number of local overlapping groups derived from global partitioning.
Similarity matrices	M	Similarity matrix of size $ P \times P $ used for clustering path candidates.
	$M_{i,j}$	Similarity score between paths p_i and p_j , based on overlapping point sets.
Path search parameters	D	Set of step lengths used in adaptive path search.
	d	A specific step length in D , defining the search radius at each iteration.
	Θ	Set of open angles defining directional constraints for path search.
	θ	A specific open angle in Θ , constraining the cone of search direction.
Combination selection sets	$\mathcal{P}'_i$	Candidate path set (reduced via clustering) for the i th local overlapping group, size $K \times N_i$.
	S'_i	Point cloud-based surrounding point combinations corresponding to $\mathcal{P}'_i$.
	$\mathcal{P}'^{j,i}$	The j th path combination in the i th local group.
	$S'^{j,i}$	Surrounding point set corresponding to $\mathcal{P}'^{j,i}$.

$5\bar{d}$, determined using a KDTree-based ball query search (Goldberger et al., 2004). Since we know each local tangent plane is centered on a known path node, the cluster containing that node is identified as the primary root cluster (Fig. 4d). For each identified primary root cluster, a circle is fitted to estimate its local radius.

Repeating the tangent plane fitting and clustering process along the entire path yields a continuous sequence of clusters delineating the primary root, while lateral roots remain scattered across different tangent slices (Fig. 4e). At each cross-section, a circle is fitted to estimate the local radius of the primary root, and these radii are recorded to

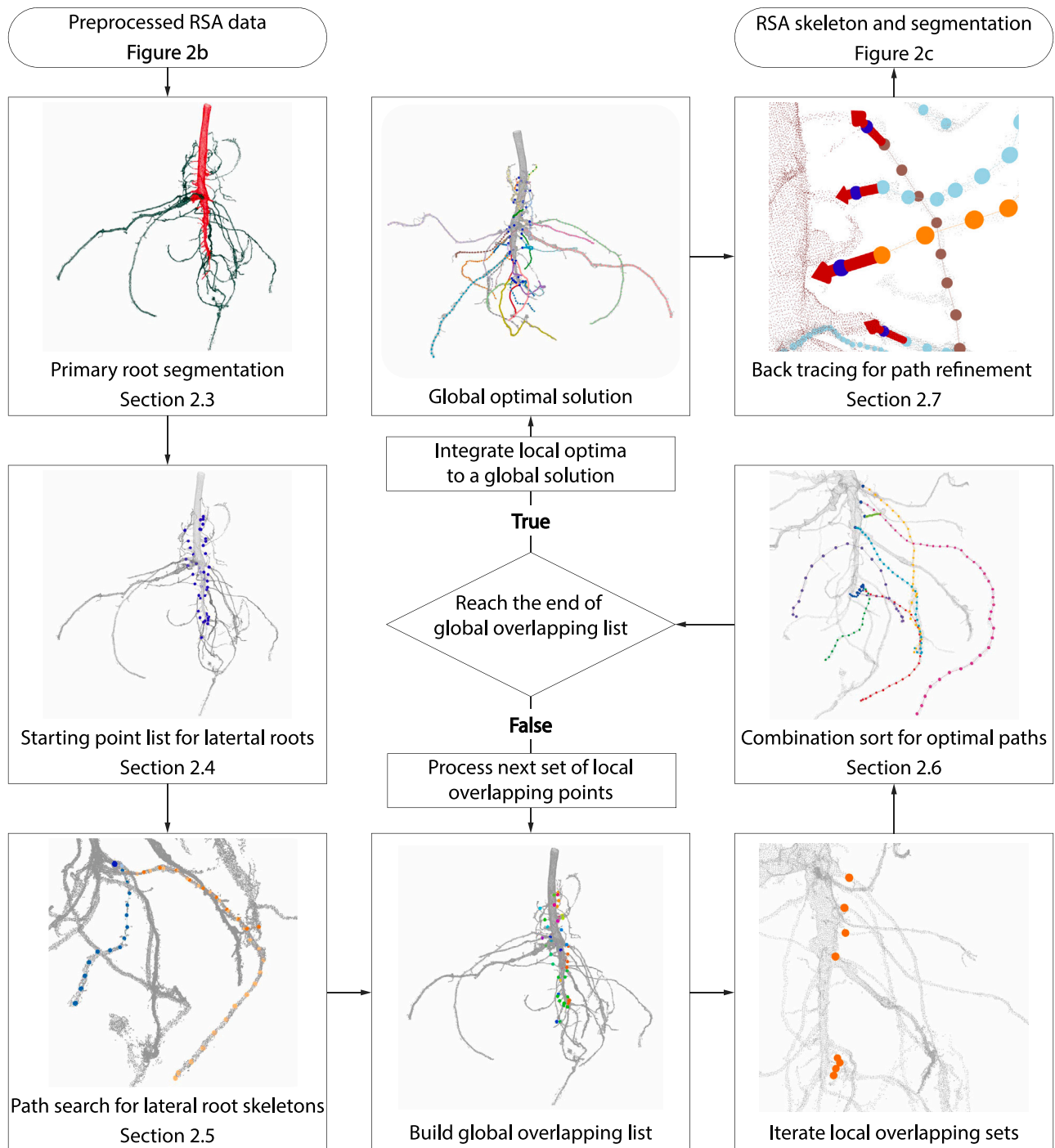


Fig. 3. Geometric skeletonization workflow. The process begins with normalized RSA data as input. First, the coarse primary root is segmented, and starting points for lateral roots are identified. From each starting point, a path searching method is applied, followed by a scoring mechanism to determine the best possible combination of paths, including those with potential overlaps. A back-tracing method is then employed to refine the identified paths. The final output is a set of geometric RSA skeletons. Detailed explanations of each step are provided in subsequent sections.

form a lookup table describing how root thickness varies with depth. To improve reliability, outlier radii are removed by excluding values that deviate excessively from the mean (e.g., beyond three standard deviations), and the remaining data are smoothed using a Savitzky–Golay filter (Savitzky and Golay, 1964) to mitigate abrupt fluctuations (Fig. 4f). These radius values are then used to define spherical query regions around each path node to further refine the primary root segmentation (Fig. 4g). Although small portions of lateral roots may still be retained, preserving these fragments aids in detecting lateral root starting points for downstream analyses.

Once the primary root point cloud is isolated, we apply a Laplacian-based skeletonization algorithm (Cao et al., 2010) to extract its medial axis (Fig. 4h). Minor lateral branches may remain in the skeleton; therefore, the Dijkstra’s shortest-path algorithm is reapplied to finalize the primary root skeleton path. Finally, we reconstruct a smooth mesh of the primary root for downstream analyses (Section 2.7). We assume that the diameter of the primary root gradually tapers from the shoot-root interface to the tip — a pattern typical of dicotyledonous plants like soybean, but potentially different in monocot species such as maize, rice, or wheat. Referring to the previously established radius

Algorithm 1 COMPUTETANGENTS

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1: Input:  $\text{path} \in \mathbb{R}^{n \times 3}$ , smoothing parameter  $\sigma$ 
2: Output:  $\text{tangents} \in \mathbb{R}^{n \times 3}$ 
3:  $\text{tangents} \leftarrow \text{COMPUTEGRADIENT}(\text{path})$ 
4: for  $i \in \{1, 2, 3\}$  do ▷ For x, y, and z coordinates
5:    $\text{tangents}[:, i] \leftarrow \text{GAUSSIANFILTER1D}(\text{tangents}[:, i], \sigma)$  ▷
   Smooth each coordinate
6:  $\text{normalized\_tangents} \leftarrow \text{tangents} / \|\text{tangents}\|$  ▷
   Normalize to unit vectors
7: return  $\text{normalized\_tangents}$ 

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lookup table, we assign an initial radius at the top and define the tip as a mathematical point with an infinitesimally small radius to simulate a realistic tapering structure. To generate the primary root mesh, we create a smooth cone along the primary root skeleton path. Tangent vectors between consecutive points determine local directions, and orthogonal circles with linearly decreasing radii from the shoot to the tip form cross-sections. Connecting these circles sequentially yields a continuous 3D primary root mesh (Fig. 4i).

2.4. Lateral root starting point localization

As illustrated in Fig. 5, we identify the starting points of the lateral roots through a boundary-based approach. Once these starting points are identified, the first-order lateral roots are inherently segmented into distinct individual entities. We begin by separating the primary root points from the lateral root points (Fig. 5a). We then focus on the lateral root points situated in close proximity to the primary root, as these boundary regions typically indicate lateral root attachment zones.

To locate these boundary points, we calculate the distance from each lateral root point to its nearest primary root point using a nearest-neighbor search. A threshold is applied to retain only those points within the closest 3% of the distance distribution. Because we apply this threshold as a percentile rather than a fixed distance value, it allows the method to be flexible and adaptable to various forms of 3D root system data, regardless of differences in scale, point density, or resolution. This percentile-based approach eliminates the need for manual tuning and ensures consistent performance across diverse datasets.

The identified boundary points are then grouped using an HDBSCAN to separate distinct lateral root attachment regions (Fig. 5b). In HDBSCAN clustering, the parameter *minimum cluster size* defines the smallest number of points required to form a valid cluster, with smaller aggregations classified as noise. Geometrically, lateral root attachment points typically form distinct, compact regions along the primary root. Setting this parameter too high may cause nearby attachment regions to merge artificially, leading to incorrect identification of multiple roots as a single cluster. Therefore, a relatively small value (e.g., 5) effectively balances noise reduction and precise separation of adjacent lateral root starting regions. The geometric centers of these clusters are computed, and these centers are recorded as the starting points of the lateral roots (Fig. 5c).

2.5. Bio-inspired lateral root skeletonization

Bionic path growth algorithm. Inspired by the continuous outward growth pattern inherent to natural root systems, we developed a bionic path growth algorithm to reconstruct the skeleton representation of lateral roots within 3D RSA data. Our approach is built upon two key assumptions: (1) lateral roots exhibit consistent outward growth from their attachment points on the primary root, and (2) the growth trajectories of lateral roots are mathematically connected and smooth. In contrast to a Kalman filter-based particle tracking approach (Mairhofer et al., 2016) – which predicts subsequent path nodes using probabilistic

models – our algorithm adopts a more computationally efficient strategy. This efficiency is crucial, as it enables the large-scale execution of extensive path searches across all lateral root starting points, facilitating the generation of a vast number of candidate paths under varying parameter settings.

For each individual path search, we will first approximate the outward growth direction of the lateral root. We begin by defining two reference points: (1) the geometric center of the lateral root attachment region, and (2) the geometric center of the corresponding attachment region on the primary root (Fig. 6a). To identify the latter, we initially expand the lateral root starting region inward along the primary root surface by a distance equivalent to $10\bar{d}$ within the point cloud. This inward expansion creates a narrow band representing the primary root side of the attachment zone, the geometric center of which is designated as the corresponding primary root attachment center. Assuming lateral roots exhibit outward growth, the initial search direction is then established from the geometric center of the primary root attachment region toward the geometric center of the lateral root attachment region (Fig. 6b). Importantly, this initial directional approximation does not require high precision, as minor inaccuracies are inherently mitigated or corrected in subsequent iterative searches or through the extensive path searching strategy employed later in our pipeline.

To determine the next node in the lateral root path, we employ a localized geometric search strategy illustrated in Fig. 6c. At each iteration, a spherical search region is established, centered at the current path base point, with a preset radius, which will be defined as the *step length* in the following discussion. To constrain the search direction and reflect the natural outward growth pattern of lateral roots, we focus only on a sector of the sphere surface. This sector is defined by a cone-shaped angular threshold, which is equivalent to the concept of open angle in the following discussion, oriented along the current path direction. We extrude this highlighted sector outward to form a virtual cylinder, limiting the candidate points to those within this cylinder. All neighboring points falling within the cylindrical search space are then retrieved. The geometric center (or density center) of these points is calculated, and this point is selected as the next node in the path.

After identifying the next node in the lateral root path, the search direction is updated to maintain alignment with the growth trajectory (Fig. 6d). Specifically, a new direction vector is computed by connecting the newly identified node to its immediate predecessor node in the path. Then, this process is iteratively repeated, updating the base and apex points, until no valid points satisfying both the step length and angular constraints can be found, indicating the termination of path extension (Fig. 6e and f).

Challenges of single-shot path search. A single-shot path search refers to applying our bionic pathfinding algorithm once for each lateral root starting point, using a uniform set of parameters for step length and angular range. However, the inherent variability in lateral root growth – such as differences in length, curvature, and spatial arrangement – presents significant challenges. Specifically, we lack prior knowledge about each lateral root's geometric properties, making it impractical to predefine universal parameters suitable for all cases. First-order lateral roots are typically densely clustered near their attachment points on the primary root and become increasingly sparse toward their tips. Applying a fixed set of step lengths and angular ranges across all lateral roots often leads to suboptimal results. Fig. 7 illustrates common challenges encountered during single-shot path searching. If the step length is too small, the algorithm fails to bridge gaps caused by data sparsity or break points, resulting in discontinuities (Fig. 7a). Conversely, overly large step lengths may cause the path to erroneously jump to adjacent lateral roots. Similarly, narrow angular ranges limit the algorithm's ability to trace sharp turns, causing premature termination, while excessively wide angular ranges increase the risk of deviating onto neighboring branches (Fig. 7b). Additionally, in regions where multiple lateral roots overlap, single-shot path searches

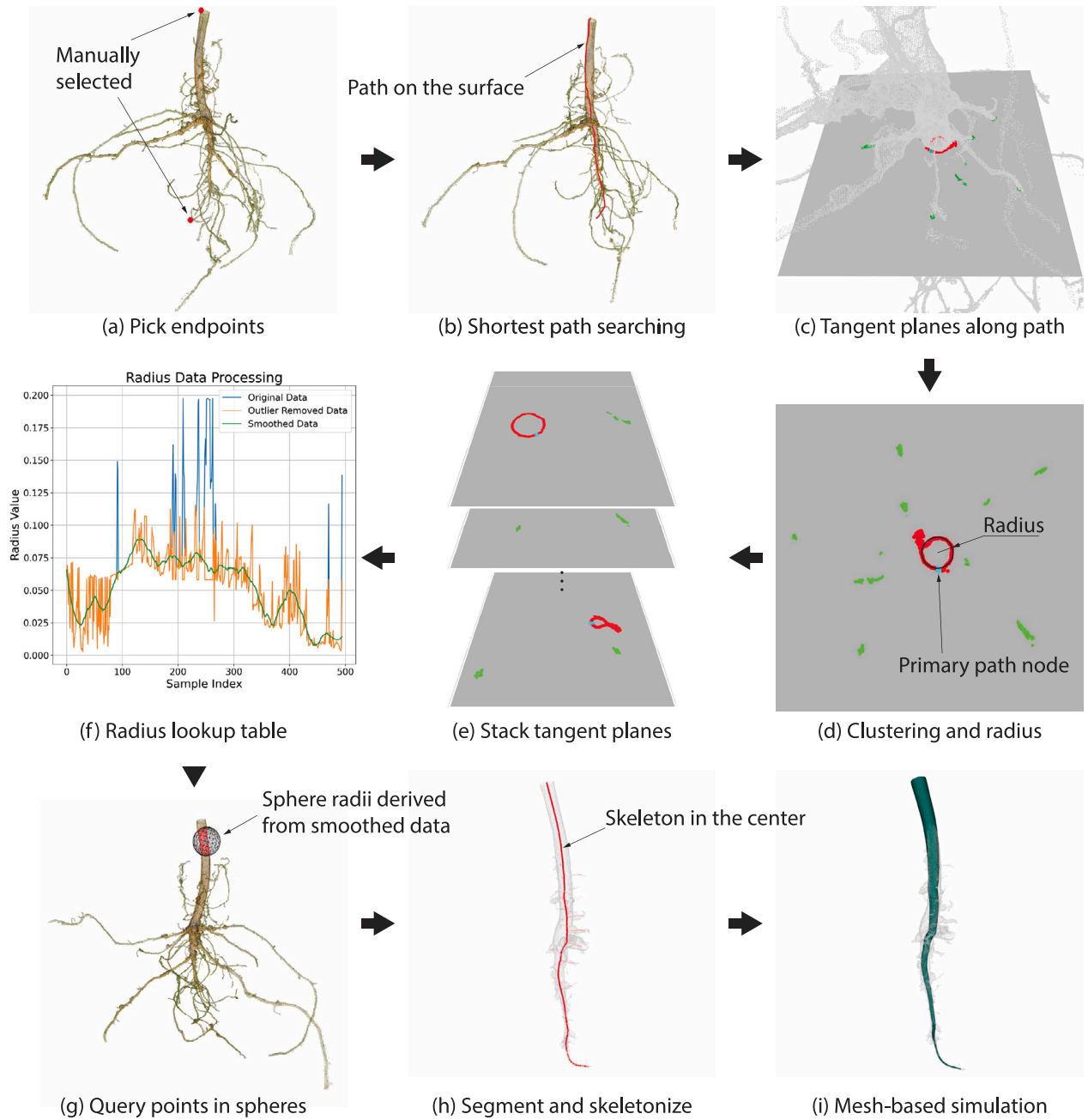


Fig. 4. Primary root point cloud segmentation, skeletonization and mesh simulation workflow. This process generates a coarse primary root point cloud, its skeleton representation, and a mesh-based model. (a) Two endpoints are manually selected, and (b) a shortest-path algorithm identifies the primary root path on the root surface. (c) Tangent planes are generated along this path, where (d) clustering on each plane identifies primary root points, and (e) circles fitted to these planes estimate radii, forming a radius lookup table. (f) This table provides sphere radii centered on each primary root path node, and (g) points within these spheres define the primary root segmentation. (h) A Laplacian-based skeletonization algorithm extracts the primary root skeleton using a shortest-path search. Finally, (i) a mesh-based model is constructed for the following back-tracing step.

initialized at different starting points may converge on the same trajectory, complicating the segmentation process and introducing ambiguity (Fig. 7c).

Adaptive pathfinding parameters. To address the challenge posed by the lack of prior knowledge regarding the geometric properties of lateral roots, we leverage the computational efficiency of our lightweight bionic path growth algorithm to perform an exhaustive search. Specifically, we generate a wide array of plausible path candidates for each lateral root by varying key parameters — step length and open angle. For every unique combination of these parameters, a distinct path candidate is produced. By introducing a comprehensive

range of parameter sets, we ensure that a large number of candidate paths are available for each lateral root starting point. This exhaustive search strategy enables us to subsequently evaluate and select the optimal combination of paths, accurately producing the skeleton of all lateral roots.

To achieve this, we introduce *adaptive search parameters* — specifically, step lengths $d_1, d_2, \dots, d_n \in D$ and open angles $\theta_1, \theta_2, \dots, \theta_n \in \Theta$. This concept is illustrated in Fig. 8a. Here, D is a set of distances that defines the search radius, where each $d \in D$ corresponds to a multiple of $\bar{d}$. Meanwhile, Θ is a set of potential angles within $[0^\circ, 60^\circ]$. Consequently, for one starting point, each combination of $d \in D$ and

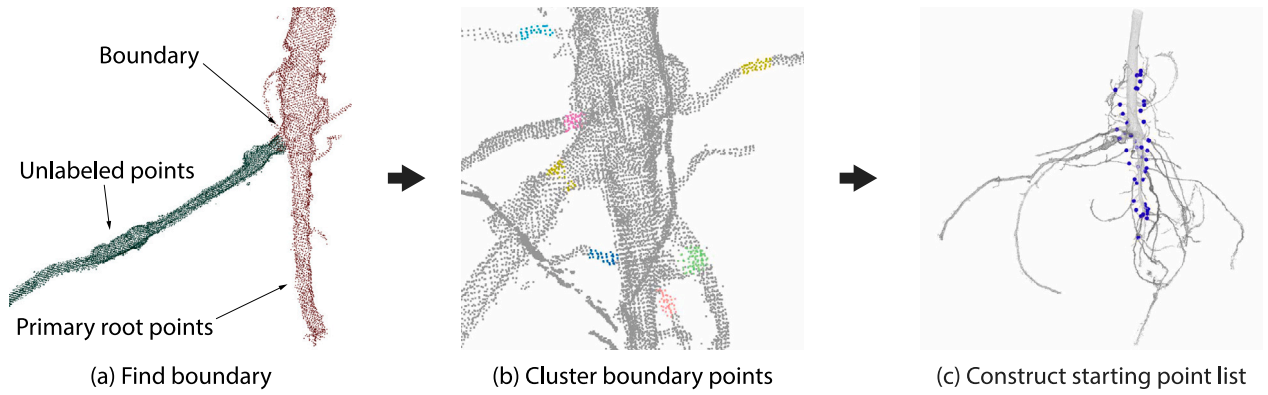


Fig. 5. Lateral root starting point list construction. (a) Boundaries are detected between the labeled primary root and unlabeled lateral root points. (b) A clustering algorithm is applied to the boundary points to separate attachment regions. (c) The geometric centers of each cluster are computed to form a list of lateral root starting points.

$\theta \in \Theta$ produces a unique candidate path p , leading to $|P| = |D| \times |\Theta|$ total path candidates. For example, $D = [2\bar{d}, 4\bar{d}, 6\bar{d}, 8\bar{d}, \dots, 20\bar{d}]$ while $\Theta = [5^\circ, 10^\circ, 15^\circ, \dots, 60^\circ]$. In this case, a single starting point would yield $|P| = 6 \times 20 = 120$ potential paths.

Extensive path searching. Thus, the adaptive pathfinding parameters serve as the strategy, while the goal is to enable extensive path search — a process visually summarized in Fig. 8b. Identifying the optimal path combination across all lateral root starting points requires evaluating every possible combination of candidate paths. Specifically, for each of the N lateral root branches or starting points, there are $|P|$ candidate paths generated through adaptive search. Therefore, the total number of possible global path combinations is $|P|^N$, which increases exponentially as the number of branches grows. This results in a computationally intractable problem, categorized as NP-hard in computational complexity theory (Cormen et al., 2022). Consequently, our objective is to reduce both N and $|P|$, making it feasible to find an optimal solution.

Path reduction by similarity-based clustering. In practice, each $|P|$ for a single starting point generates many similar paths. To mitigate this redundancy, we cluster these similar paths by constructing a similarity matrix M for all path candidates $p \in |P|$ corresponding to each starting point. Here, M is a $|P| \times |P|$ symmetric matrix whose main diagonal is composed of ones, reflecting that each path is identical to itself.

For a given path p , we define its associated set of surrounding points s , and use S to represent all such sets. The surrounding point set s for a path is determined using a ball query centered at each point along the path. Specifically, for the k th point $p[k]$ in the path, the ball query radius is set to the Euclidean distance between $p[k]$ and its preceding point $p[k-1]$, i.e., $\|p[k] - p[k-1]\|$. This adaptive radius ensures the query scale reflects local path geometry. Let p_i and p_j be two paths with surrounding point sets s_i and s_j , respectively. We compute their similarity based on the normalized size of the intersection of s_i and s_j :

$$M_{i,j} = M_{j,i} = \frac{1}{2} \left(\frac{|s_i \cap s_j|}{|s_i|} + \frac{|s_i \cap s_j|}{|s_j|} \right). \quad (1)$$

We then apply spectral clustering to M in order to group similar paths into K clusters (Von Luxburg, 2007). Within each cluster, we select the path that has the largest number of surrounding points, as this maximizes the coverage of our skeleton representation for each lateral root. By consolidating similar paths, we greatly reduce the overall search space for P , making it feasible to find an optimal combination of paths that accurately represents the lateral root structure. We illustrate a path reduction result by clustering in Fig. 8c, while the steps of the similarity-based clustering of paths method are detailed in Algorithm 2.

Algorithm 2 SIMILARITY-BASED CLUSTERING OF PATHS

```

1: Input:  $|P|$  (number of paths),  $S$  (array of path surrounding points),  $K$  (desired number of clusters)
2: Output: bestPathsInCluster  $\triangleright$  Map from each cluster to its best path
3:  $M \leftarrow \text{zeros}(|P| \times |P|)$   $\triangleright$  Initialize an  $|P| \times |P|$  matrix of zeros
4: for  $i \leftarrow 0$  to  $|P| - 1$  do
5:   for  $j \leftarrow i + 1$  to  $|P| - 1$  do
6:      $\text{ratio} \leftarrow \text{CALCOVERLAP}(S[i], S[j])$   $\triangleright$  Eq. (1)
7:      $M[i, j] \leftarrow \text{ratio}$ 
8:      $M[j, i] \leftarrow \text{ratio}$ 
9:  $\text{spectral} \leftarrow \text{SPECTRALCLUSTERING}(n\_clusters = K)$ 
10:  $\text{clusters} \leftarrow \text{spectral.fit\_predict}(M)$ 
11:  $\text{bestPathsInCluster} \leftarrow \{\}$   $\triangleright$  Final map of best path per cluster
12: for each  $\text{clusterId}$  in  $\text{clusters}$  do
13:    $\text{bestPathsInCluster}[\text{clusterId}] \leftarrow \text{SELECTMAXSURROUNDINGPOINTS}(\text{clusters}[\text{clusterId}])$ 
14: return  $\text{bestPathsInCluster}$ 

```

2.6. Combination sort for local optimal paths

As illustrated in Algorithm 2, we reduce the number of candidate paths $|P|$ for each lateral root starting point to a constant K via spectral clustering. In our experiments, we set $K = 4$, effectively reducing the path set P to $P' \subset P$, where $|P'| = K$. This value provides a practical trade-off between computational efficiency and candidate diversity, though it can be adjusted based on the available computational resources — larger values of K allow for a more comprehensive search space at the cost of increased processing time. From each starting point, these K representative paths form the refined set P' , from which the most suitable path can be selected. Consequently, the combinatorial complexity of selecting one path per branch across all N starting points becomes $\mathcal{O}(K^N)$. Given that N could be a large number in densely branched root systems, even with reduced P' , the resulting computational burden remains substantial and necessitates further optimization.

Local overlapping list. To address this, we further reduce the exponential N before searching for the optimal path combination. Note that distinct branches often show clear geometric differences (e.g., they are far apart in Euclidean space and do not intersect) and therefore do not conflict when identifying the global optimum. Hence, we group these non-overlapping branches into separate sets, effectively splitting N into l mutually independent local subsets $N_1, N_2, \dots, N_l$. We

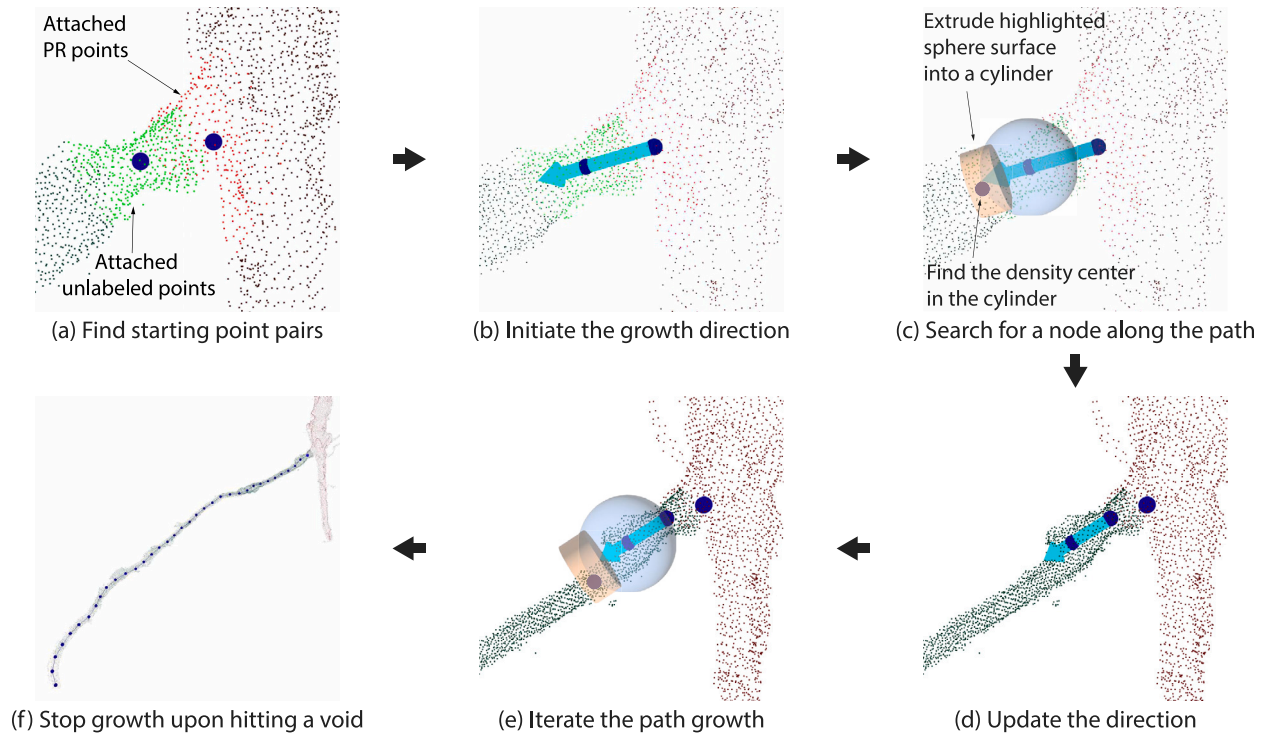


Fig. 6. Lateral root path searching demonstration. After identifying the lateral root starting points: (a) The attached points on the primary root and lateral roots are located, and their geometric centers are used to define starting point pairs. (b) Using these initial pairs, the initial growth direction is established. (c) The next path node is determined within a predefined range. (d) The growth direction is updated after identifying the new path node. (e) This path search method is iteratively repeated to identify subsequent nodes. (f) The process continues until the path search encounters a void.

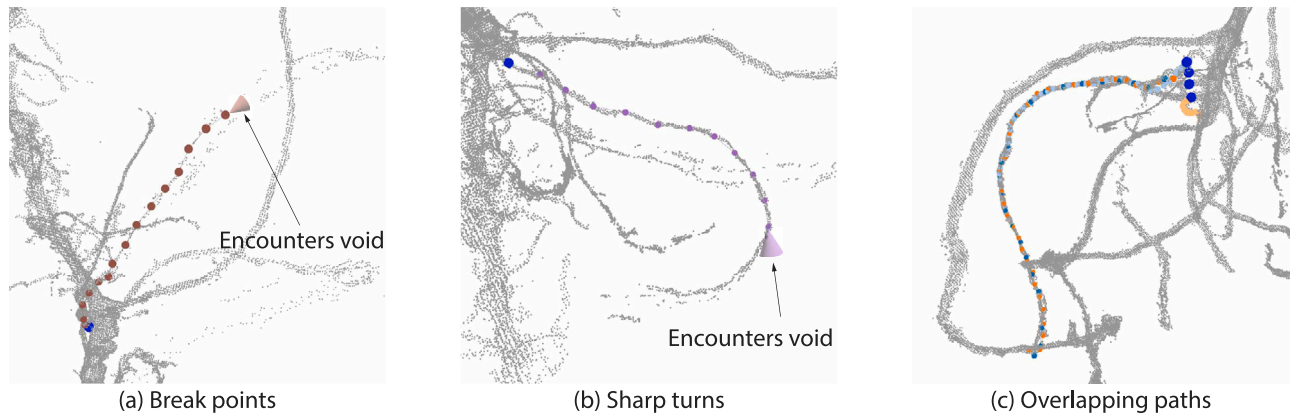


Fig. 7. Challenges in the dataset. Single path search might not be sufficient when encountering (a) break points, (b) sharp turns, and (c) overlapping scenarios, as these complexities require more robust strategies to ensure accurate pathfinding for skeletonizations.

denote the resulting local path combination subsets (after clustering reduction) as $P'_1, P'_2, \dots, P'_i$, where each P'_i contains $K \times N_i$ candidate paths and K_i^N path combinations. Fig. 9a illustrates one such P'_i example of a local overlapping group. Consequently, the computational complexity is reduced from $\mathcal{O}(K^N)$ to $\mathcal{O}(K^{N_1} + K^{N_2} + \dots + K^{N_i})$. Although the theoretical complexity remains exponential, the umbrella-like spatial divergence of root systems makes it highly unlikely for a large number of overlapping branches to accumulate in a single group N_i . This characteristic enables practical efficiency gains, making the approach highly scalable and effective for complex 3D RSA data. For splitting N , we compare the candidate paths P' for all N branches by their Euclidean distances and angles, thereby determining whether two

branches overlap. For details on this comparison process, please refer to the pseudocode in Algorithm 3.

Combination sort. Once the overall search space has been partitioned into local subsets P'_i , our goal is to systematically evaluate and rank all possible path combinations within each subset to identify the most suitable solution. We proceed under two key assumptions: (1) after removing the primary root point cloud, each remaining point belongs to one and only one first-order lateral root; and (2) overlaps between lateral roots are typically cross-like intersections confined to small, localized regions, rather than extensive tangling across large areas. Guided by these assumptions, we aim to maximize the coverage of the lateral root skeletons – ensuring that they collectively represent as

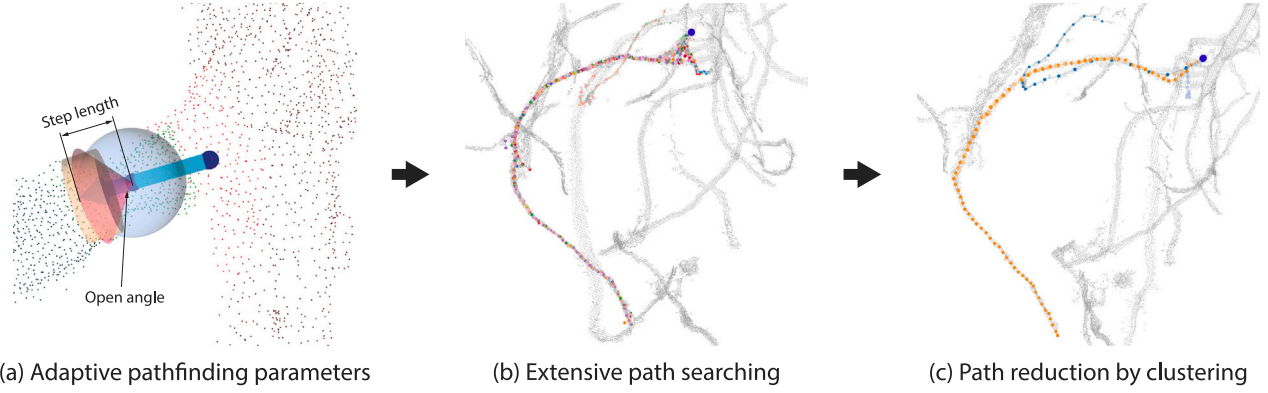


Fig. 8. Brute-force path search from a single starting point. (a) By varying key search parameters (e.g., open angle and step length), multiple candidate paths are considered. (b) This extensive search generates a large set of potential lateral root paths. (c) To improve computational efficiency, a spectral clustering method is applied to identify and remove redundant paths with similar patterns.

Algorithm 3 CHECKOVERLAP

```

1: Input: path1, path2
2: Output: True if paths overlap, False otherwise
3: distance  $\leftarrow$   $\|path1 - path2\|$   $\triangleright$  Euclidean distance
4: if distance  $> d_{threshold}$  then
5:   return False
6: dot_product  $\leftarrow$  path1  $\cdot$  path2
7: norm_v1  $\leftarrow$   $\|path1\|$ 
8: norm_v2  $\leftarrow$   $\|path2\|$ 
9: if norm_v1 == 0 or norm_v2 == 0 then
10:  return False  $\triangleright$  Handle zero vectors
11:  $\cos(\theta) \leftarrow \frac{dot\_product}{norm\_v1 \times norm\_v2}$ 
12: if  $\cos(\theta) > \cos(\frac{\pi}{4})$  or  $\cos(\theta) < \cos(\frac{3\pi}{4})$  then
13:  return True  $\triangleright$  Within range  $[-45^\circ, 45^\circ]$ 
14: else
15:  return False

```

many points in the RSA point cloud as possible – while simultaneously minimizing overlaps. Therefore, our evaluation should operate at the point cloud level rather than solely at the skeleton level.

For each candidate path p , we determine its associated set of surrounding points s using a same ball query approach, employed in Eq. (1), ensuring that the local geometry around each path segment is well captured. Consequently, each local path combination subset $\mathcal{P}'_i$, containing K^{N_i} possible path combinations, can be extended to its corresponding point set S'_i , which comprises all possible combinations of surrounding points derived from $\mathcal{P}'_i$. Specifically, we denote each individual path combination within $\mathcal{P}'_i$ as $\mathcal{P}'_i^1, \mathcal{P}'_i^2, \dots, \mathcal{P}'_i^{K^{N_i}}$, and the associated surrounding point combinations as $S'^1_i, S'^2_i, \dots, S'^{K^{N_i}}_i$. Our objective then becomes ranking all combinations in S'_i according to a scoring criterion that balances coverage and overlap. The optimal solution corresponds to the combination S'^j_i with the highest score, and its associated path combination $\mathcal{P}'^j_i$ is selected as the best local configuration.

To balance the objectives of maximizing path coverage while minimizing overlap, we designed a scoring criterion and summarized it in Algorithm 4. For computational efficiency, all points in the 3D RSA data are indexed as unique integers. Given a specific combination of paths S'^j_i in a local subset $\mathcal{P}'_i$, we iterate through each path segment and collect the indices of its surrounding points, appending them to an aggregated occurrences list. Then, we compute the frequency of each unique index to capture how many times a particular point appears across different paths in the combination. Points that are exclusively

covered by a single path (i.e., the number of occurrences equals one) are considered desirable and contribute positively to the score, promoting a wide coverage of the root system. Conversely, if a point is shared by multiple paths, it incurs a penalty to discourage overlapping coverages.

Formally, for each unique point index, we increment the score by one for the first occurrence and subtract one for every additional occurrence. This ensures that each point's contribution is maximized only when it belongs uniquely to a single path while penalizing overlaps. After all occurrences are processed, the final score reflects the trade-off between point coverage and overlaps for path surrounding point of each combination. The combination of surrounding points, along with its corresponding set of lateral root paths, yielding the highest score is selected as the optimal configuration.

Algorithm 4 COMPUTEScore

```

1: Input: A path surrounding point combination  $S'^j_i$  of local overlapping lateral roots  $\mathcal{P}'_i$ .
2: Output: A score based on the number of points with single occurrence (single occupation) minus the number of points with multiple occurrences (overlaps).
3: Initialize occurrences as an empty list
4: for segment in  $S'^j_i$  do
5:   for index in segment do  $\triangleright$  Each point has a unique index
6:     Append index to occurrences
7: Initialize occurrence_counts as the frequency of each point in occurrences
8: Initialize score to 0
9: for count in occurrence_counts do
10:  if count == 1 then
11:    Increment score by 1  $\triangleright$  +1 for a single occurrence
12:  else
13:    Increment score by 1 - count  $\triangleright$  +1 for the first occurrence, -1 for each additional occurrence
14: return score

```

An example of an optimal local path combination is presented in Fig. 9b, highlighting the effectiveness of our scoring and sorting strategy in resolving overlaps and maximizing coverage. Once the optimal local combinations for all subsets $\mathcal{P}'_i$ are identified, these local solutions are aggregated to construct the globally optimal path configuration encompassing all first-order lateral roots. The final integrated result is illustrated in Fig. 9c.

2.7. Back tracing refinement

Counterintuitively, a precise primary root segmentation might introduce inaccuracies to our skeletonization algorithm. If the segmentation

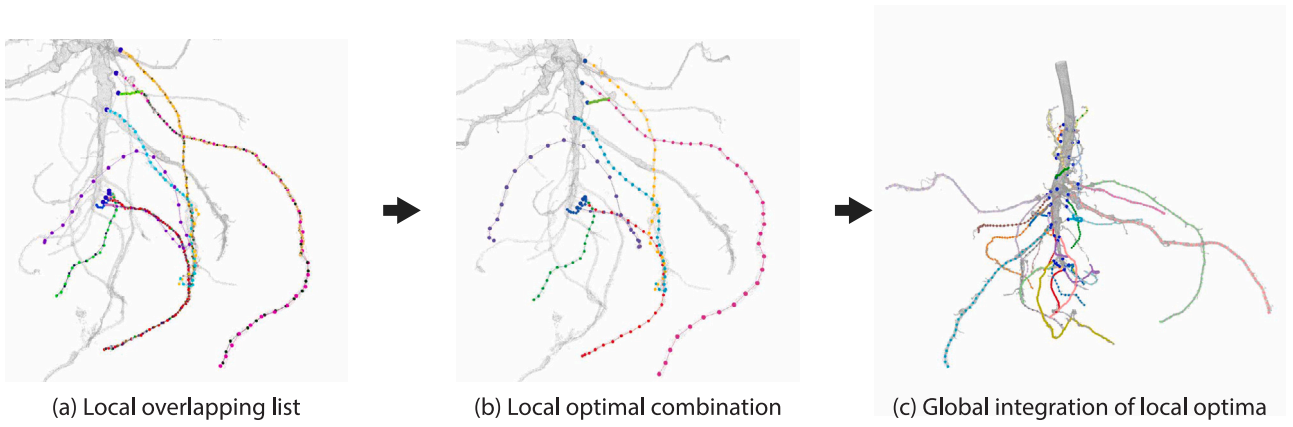


Fig. 9. Finding the optimal lateral root path combination. After obtaining reduced path candidates from each starting point, (a) potential overlapping groups are identified, (b) the best path combination is selected within each group, and (c) all local optima are integrated to achieve a globally optimal solution.

is highly precise, resulting in a perfectly delineated primary root, neighbor lateral roots are very likely to share a same attachment region. This can cause the algorithm to misidentify multiple lateral root starting points as a single one. Even if the algorithm correctly distinguishes them as separate lateral roots, their paths may interfere with each other during the bionic path growth stage. The ideal approach is to extend the primary root segmentation slightly to include the attachment areas of lateral roots. This ensures clearer distinctions between different lateral roots, as they typically diverge in their growth directions further from the attachment site. However, incorporating lateral root attachment areas in the primary root segmentation can introduce minor inaccuracies in the skeleton representation, particularly in the joint regions, where fine structures might be lost. To address this, we introduce a back-tracing method, where pathfinding proceeds inward rather than outward. This process is illustrated in Fig. 10.

The input to the back-tracing procedure consists of the globally sorted optimal paths identified in the previous stage (Fig. 10a). Unlike the initial outward search, we now focus on the inward direction by selecting the first two nodes of each path – those closest to the primary root – as the starting points to initialize the inward search vector (Fig. 10b). The same path growth algorithm used during outward searching is employed here to iteratively trace the lateral root paths backward (Fig. 10c). A key component of this back-tracing procedure is the definition of a robust stopping criterion. In contrast to the outward search, which terminates naturally when no neighboring points are found, inward tracing requires a geometric condition to determine when the search has reached the connection site between the lateral and primary roots. To facilitate this, a primary root mesh is generated following the segmentation of the primary root. During each iteration of the back-tracing process, a detection ray is cast from the current point in the search direction, with a length equal to the back-tracing step size (Fig. 10d). If the detection ray intersects the primary root mesh, it signals that the traced path has reached the root junction, and the process is terminated (Fig. 10e). Otherwise, the search continues iteratively until this condition is satisfied. An example of a refined lateral root skeleton, achieved through the back-tracing process, is shown in Fig. 10h. This refinement enhances the accuracy of the skeleton representation, thereby improving the precision of subsequent trait extraction and segmented root system analysis.

2.8. Trait extraction

Before trait computation, all RSA data are rescaled to their actual size using the scaling factor applied during preprocessing. The number of lateral roots is determined by either summing the number of starting points identified at the primary-lateral root junctions or by counting the number of segmented lateral root paths in the final skeleton. The

root length is calculated as the cumulative sum of distances between consecutive nodes along each skeleton path.

Roots exhibit complex 3D architectures, and individual lateral roots often follow highly curved spatial trajectories, making direct 3D angle measurements unreliable. To obtain a meaningful 2D angular representation, we first project the root system from 3D to 2D using Principal Component Analysis (PCA) (Pearson, 1901). The primary root PCA defines the reference axis, while each lateral root PCA characterizes the dominant orientation of individual lateral roots. The branching angle of each lateral root is computed as the angle between the primary root's PCA vector and the lateral root's PCA vector.

Accurately estimating lateral root surface area and volume is essential for genome-wide association studies (GWAS) and quantitative trait locus (QTL) mapping, as these traits provide critical insights into RSA and its influence on water and nutrient uptake, stress tolerance, and overall plant productivity (Torkamaneh and Belzile, 2022; Topp et al., 2013). However, precise quantification of these traits from 3D data is challenging due to the irregular and tapering nature of root structures. To overcome this, we reconstructed tube-like surface meshes for each individual lateral root based on its skeletonized centerline, as shown in Fig. 11a. Given that lateral roots exhibit non-uniform diameters along their length, we first divided each path into fixed-length chunks and estimated the local radius for each chunk by analyzing the perpendicular distances from surrounding point cloud data to the corresponding path segment. The local orientation of each chunk was determined using PCA, which allowed circular cross-sections to be fit perpendicular to the dominant direction of each segment. The resulting chunk-wise radii were then processed to enforce a monotonic non-increasing trend from base to tip, reflecting natural root tapering, followed by Gaussian smoothing to reduce local fluctuations. These refined radii were interpolated along the entire path to assign per-node radius values, and smooth tube-like meshes were generated by connecting adjacent circular cross-sections along the skeleton trajectory.

Following tube mesh reconstruction, we calculated the surface area and volume of each lateral root by analytically integrating over the interpolated radii and segment lengths. Surface area was estimated by modeling each pair of adjacent circular cross-sections as a truncated cone. Specifically, for each pair of neighboring nodes along the skeleton path, we computed the segment length and the corresponding radii r_1 and r_2 . The slant height of the truncated cone was then calculated using the Euclidean distance between path points and the radial difference, and the lateral surface area of each segment was computed as

$$A = \pi(r_1 + r_2)\sqrt{(r_2 - r_1)^2 + \text{length}^2}. \quad (2)$$

Summing the surface areas across all segments yielded the total surface area of the root. For volume estimation, we adopted the analytical

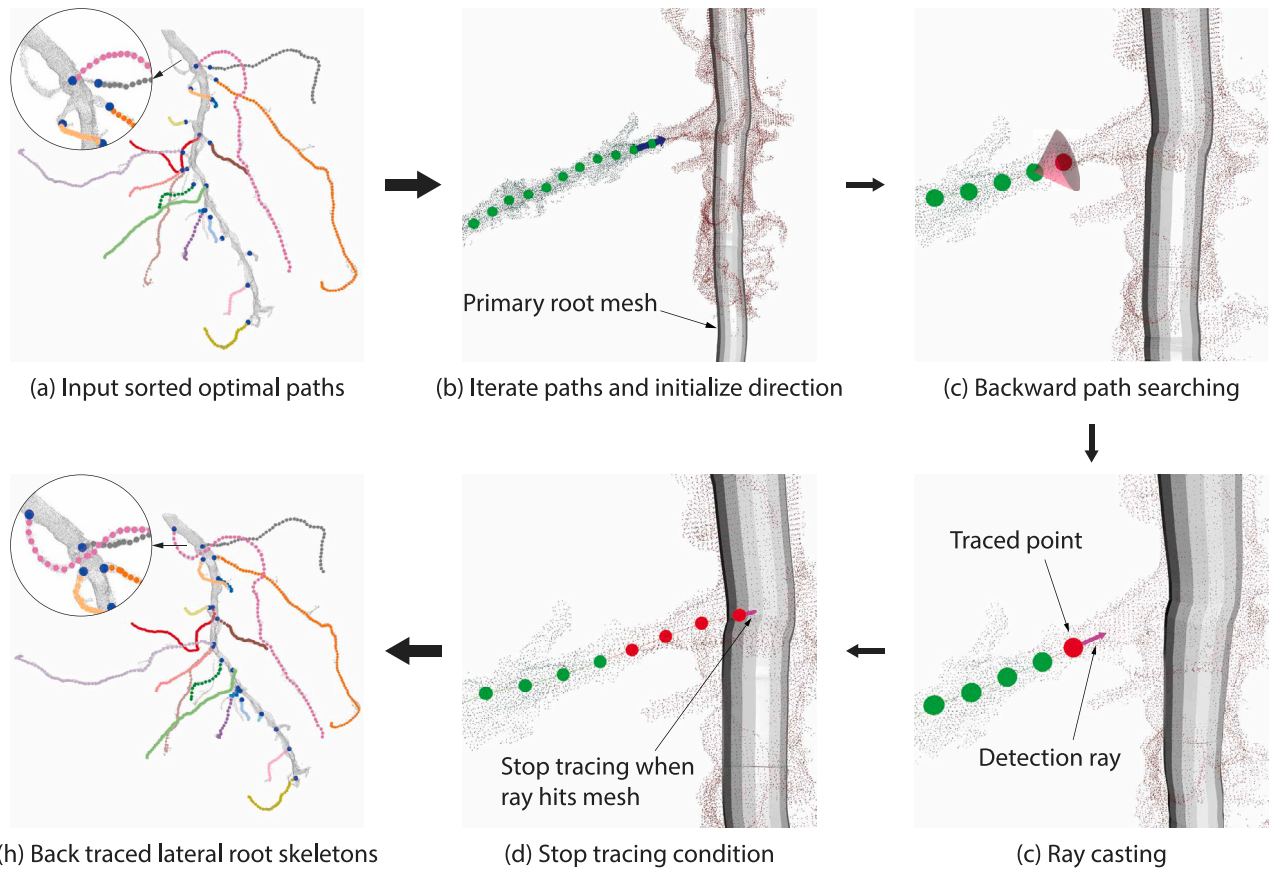


Fig. 10. Back-tracing procedure to refine lateral root skeletons. (a) Input sorted paths. (b) Initialize inward direction. (c) Perform backward path searching. (d) Cast detection ray. (e) Stop tracing upon ray–mesh collision. (h) Resulting refined skeletons with improved connectivity to the primary root.

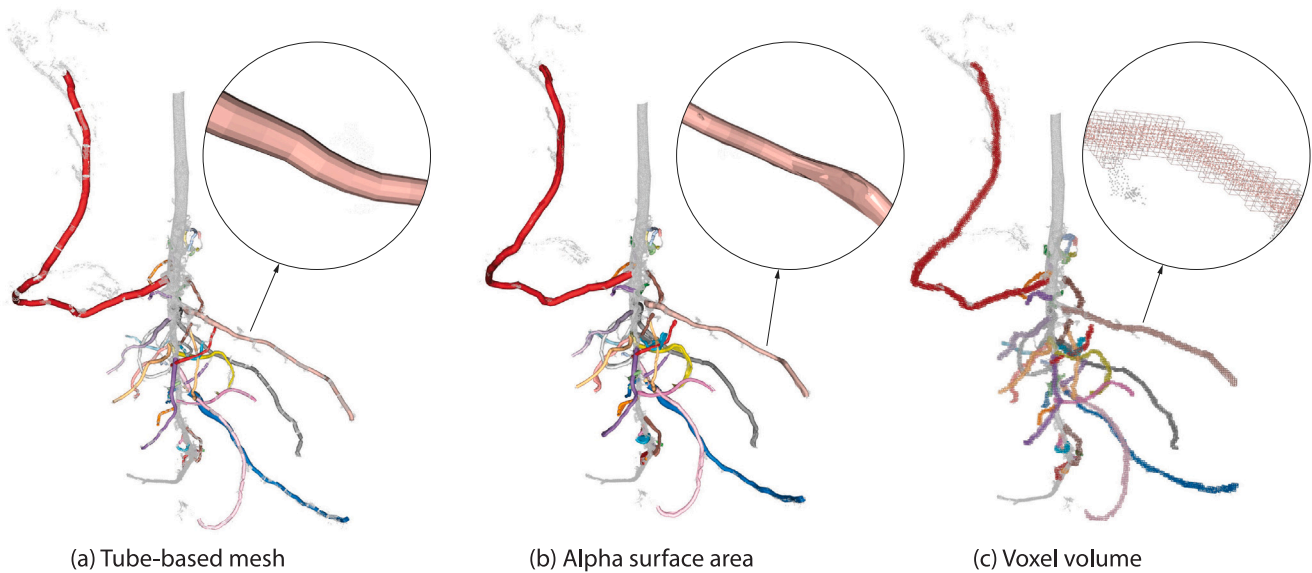


Fig. 11. Reconstructed tubes were used for surface area and volume calculation of individual lateral roots. (a) Reconstructed tube meshes demonstrate geometry surrounding individual root paths for estimating surface area and volume. (b) Alpha surface reconstruction was used to cross-validate surface area estimations by generating a fitted boundary around root segments. (c) A voxel-based method was employed to cross-validate volume calculations by quantifying occupied voxel space.

expression for the volume of a frustum: each segment's volume was calculated using the formula

$$V = \frac{1}{3}\pi h(r_1^2 + r_1r_2 + r_2^2), \quad (3)$$

where h is the segment length and r_1, r_2 are the radii of the segment's endpoints. The total root volume was then obtained by summing the volumes of all frustum-shaped segments.

To validate the accuracy of these mesh-based calculations, we employed two complementary approaches. First, alpha shape reconstruction was used to estimate the surface area of each lateral root (Fig. 11b). Alpha shapes generalize the convex hull by enabling tight, non-convex enclosures of scattered points based on a parameter α (i.e. 0.1) that controls the level of detail in the resulting surface (Edelsbrunner et al., 1983). The surface area derived from alpha shapes serves as an independent geometric estimate that reflects the spatial distribution of the point cloud. Second, we used a voxel-based method to estimate the volume of each root (Fig. 11c). In this approach, the root system was voxelized into a 3D grid at $5\bar{d}$, and the total volume was calculated by counting the number of occupied voxels and multiplying by the unit voxel volume. This provided a discrete, geometry-agnostic reference for validating the tube-derived volume estimates.

2.9. Computing hardware and environment

All experiments were conducted on a dedicated workstation equipped with a 12th Gen Intel® Core™ i7-12700H CPU, 64 GB of RAM, and an NVIDIA GeForce RTX 3080 Ti GPU. This setup provided sufficient computational resources to support intensive 3D point cloud processing, root segmentation, and skeletonization tasks.

The implementation was developed in Python and relied on a combination of scientific and machine learning libraries. Key packages include *open3d* and *matplotlib* for 3D processing and visualization; *numpy* and *scipy* for numerical computation; *hdbscan*, *scikit-learn*, and *networkx* for clustering and graph analysis; *scipy.optimize*, *savgol_filter*, and *scipy.ndimage* for optimization and signal processing; and other tools such as *pickle*, *tqdm*, *threading*, *concurrent.futures*, and Python's built-in libraries for utility functions.

3. Results and discussion

We have discussed all hyperparameters in the Methods section. Below, we provide the exact values used in our experiments and elaborate on their effects during tuning: (1) The *minimum cluster size* in HDBSCAN for primary root identification was set to 5. Increasing this value may result in missing small yet valid clusters that represent parts of the primary root, whereas decreasing it might introduce noise by accepting spurious small clusters. (2) The *minimum cluster size* in HDBSCAN for lateral root starting point localization was set to 4. A higher value may cause some lateral root starting points to be ignored due to their smaller point count, while a lower value may lead to false positives due to noise. (3) The *adaptive step length* for bio-inspired path tracing was set to $[2d, 4d, 6d, 8d, 10d]$. Increasing the number of step lengths can improve robustness in cases with discontinuous or noisy point clouds, but it also increases runtime and may reduce computational efficiency. (4) The *open angle threshold* was set to $\cos(\theta) = [0.25, 0.5, 0.7]$, corresponding approximately to angles of 75° , 60° , and 45° . Similar to step length, adding more angular thresholds increases path diversity and robustness, but at the cost of longer runtimes due to more candidate paths. (5) The *number of clusters* K in similarity-based clustering during the path selection phase was set to 4. Reducing K decreases the number of candidate paths and may reduce robustness, particularly during the lateral growth stage. Conversely, increasing K may improve robustness in highly branched cases but leads to exponential growth in runtime. (6) A *neighbor radius* of $4d$ was used to extract skeleton-surrounding points for lateral root segmentation. A larger radius better accommodates thicker roots but may introduce surrounding noise. A smaller radius ensures cleaner segmentation but may result in overly thin or incomplete lateral root segments.

3.1. 3D skeletonization comparison

Laplacian-based skeletonization has been widely adopted in 3D plant phenotyping. Numerous studies have employed this method for species such as sorghum (Gaillard et al., 2020), cotton (Sun et al., 2021), tomato and tobacco (Ziamtsov et al., 2021), apple trees (Qiu et al., 2024), and cherry trees (Meyer et al., 2023). However, these plant structures are relatively simple compared to the intricate architecture of soybean root systems. Fig. 12 presents a comparison between a fine-tuned Laplacian-based skeletonization and our method, highlighting key limitations of the Laplacian approach and advantages of our framework. The average runtime for each step of our pipeline, broken down by data collection date, is summarized in Table 2, demonstrating the computational efficiency of our parallelized implementation.

Limitations of Laplacian skeletonization. Laplacian-based methods are well-suited to smooth, manifold-like geometries but face significant challenges when applied to the non-manifold nature of RSA, where intersecting and overlapping lateral roots are common. Additionally, RSA point clouds typically exhibit substantial variability in point density, with sparse sampling at lateral root tips and dense clustering at primary-lateral junctions. Because Laplacian skeletonization applies global edge contractions and vertex movements uniformly – without adapting to local density – it often leads to discontinuities in sparse regions and unwanted fusions in dense areas. Furthermore, the iterative smoothing inherent in Laplacian contraction risks excessive shrinkage, which may eliminate fine lateral root details if contraction forces are too strong. Conversely, weaker forces can induce artificial ring formations, further compromising skeleton fidelity. Another drawback emerges in data derived from multi-view stereo imaging: environmental turbulence frequently introduces noise, particularly around lateral roots. While Laplacian methods are theoretically robust to noise, they struggle to distinguish genuine root structures from background artifacts. Critically, Laplacian skeletonization lacks intrinsic support for individual root segmentation, often merging closely spaced lateral roots or failing to distinguish separate branches – hindering its applicability for precise phenotyping tasks.

Advantages of the bionic root parser. In contrast, our bionic root system parser adopts a bottom-up, biologically inspired approach designated to complex root architectures like soybean RSA, and potentially extensible to other tree-like or root-like systems. Rather than simplifying the root structure via global contraction, our method dynamically reconstructs the skeleton by tracing natural growth trajectories. This ensures preservation of both primary and lateral root structures without over-smoothing or shrinkage. Leveraging adaptive pathfinding parameters, the parser adjusts step lengths and angular search ranges based on local point densities – preventing breakages in sparse areas and avoiding topological distortions in dense regions. A key strength of our method lies in its robustness to non-manifold configurations and overlapping lateral roots: by treating each lateral root as an independent path, erroneous merging is avoided, and individual topologies are retained. Moreover, the parser's assumption of outward, smooth growth constrains the search space to biologically plausible regions, significantly reducing susceptibility to noise and outliers. Paths affected by noise are naturally eliminated during the two-tiered selection process – first, through clustering-based reduction, and second, via combination scoring and sorting – ensuring that only actual root trajectories are preserved. This robustness is demonstrated in the middle-right subfigure of Fig. 12, where our method accurately tracks the natural course of the lateral root despite noise and structural complexity.

3.2. Point cloud-wise segmentation results

Fig. 13 illustrates the output of our skeletonization framework (left) alongside the corresponding point cloud-wise segmentation (right). This segmentation provides a comprehensive breakdown of the 3D RSA, enabling not only topological analysis but also facilitating downstream

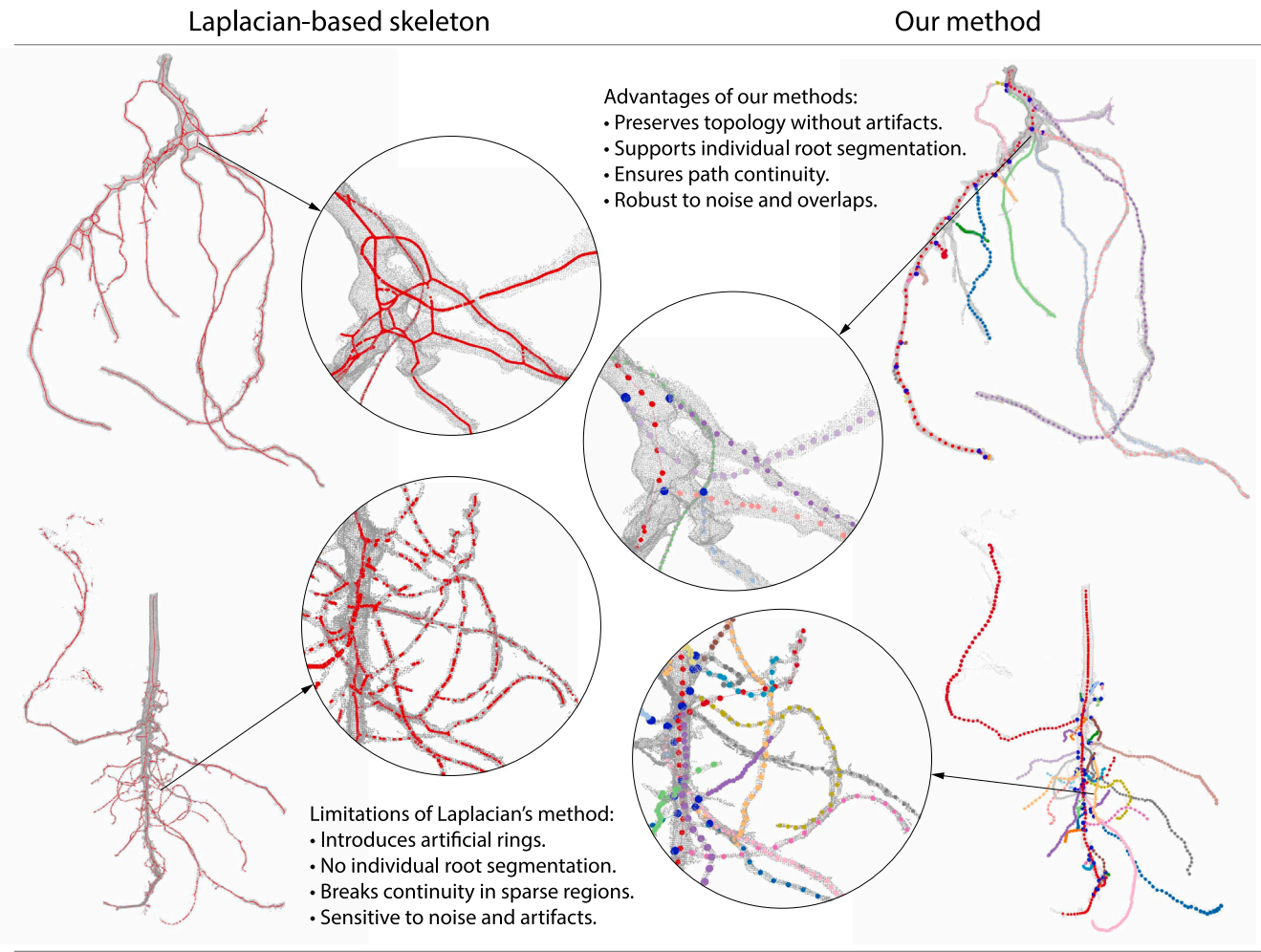


Fig. 12. Comparison between our method (right) and Laplacian-based skeletonization (left) on 3D RSA data. Our approach preserves topology, ensures continuity, segments individual lateral roots, and is robust to noise and overlaps.

Table 2
Average runtime (in seconds) for each step of the root system analysis pipeline, reported separately for the 15-day and 30-day old soybean root systems. The method was optimized using parallel computing with 8 workers and executed on a workstation with a 12th Gen Intel(R) Core(TM) i7-12700H CPU, 64 GB RAM, and an NVIDIA GeForce RTX 3080 Ti GPU.

Breakdown steps	15-day old soybean root	30-day old soybean root
Primary root identification	24.8 s	48.0 s
Lateral root starting point localization	1.8 s	2.8 s
Bio-inspired lateral root skeletonization	42.4 s	165.4 s
Combination sort for local optimal paths	55.9 s	237.4 s
Back tracing refinement	7.2 s	11.6 s
Average point size	91,958	154,313
Total Time	132.1 s	465.2 s

quantitative assessments such as individual root volume estimation, surface area measurement, and localized density analysis. Specifically, we perform a ball query around each skeleton node for lateral roots, the same method employed in Eq. (1), ensuring that nearby points are accurately grouped. For the primary root, we employ a similar strategy, querying all anchor points derived from the primary root mesh.

While the ball query method offers a robust and efficient approach for segmentation, some minor inaccuracies may arise in densely populated regions, such as the primary–lateral root junctions, where overlapping structures result in ambiguous point assignments. To mitigate this, for any points encompassed by multiple queries, we can compute their distances to all involved paths and assign each point to the nearest

path. Although slight uncertainties may still persist at these junctions, they represent only a small fraction of the total root system and have minimal influence on the accuracy of global trait calculations.

3.3. Trait analysis

We present the results of our trait analysis in Fig. 14, demonstrating the accuracy of our trait extraction pipeline and its relevance to studies of RSA. These traits are critical for soybeans as they are directly associated with nutrient uptake efficiency, drought resilience, and yield performance.

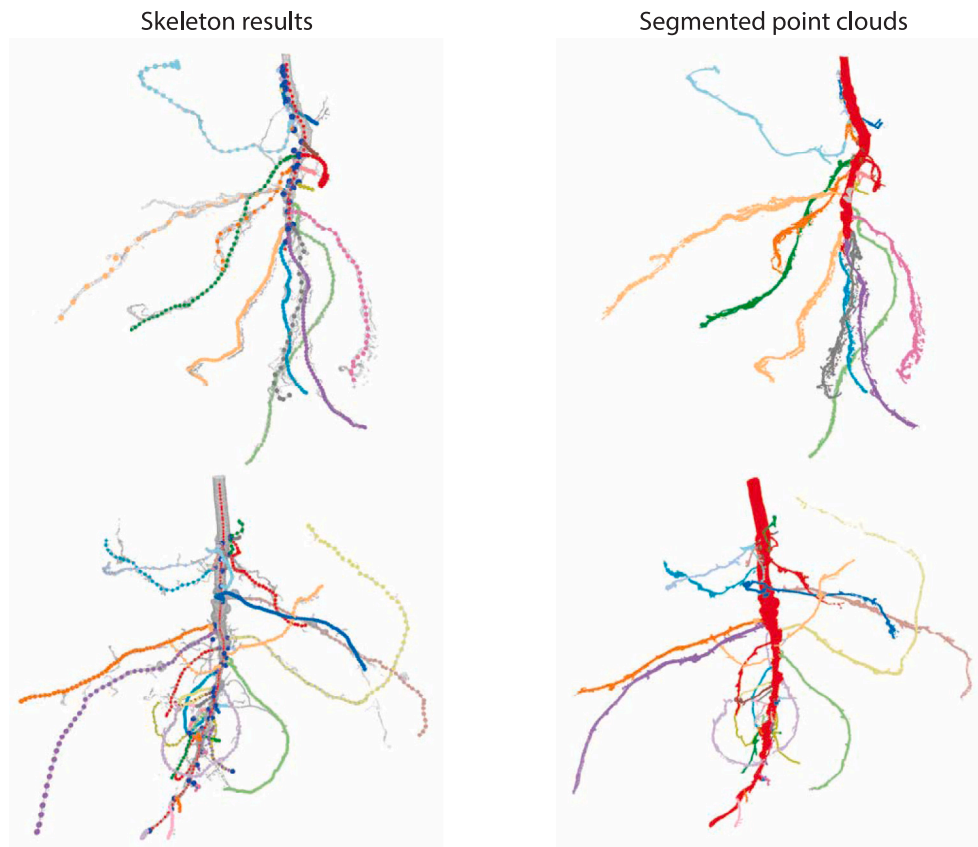


Fig. 13. Skeletonization and segmentation results from our proposed method on 3D RSA data. The left column shows the extracted skeletons, where individual root paths are assigned distinct colors. The right column displays the corresponding segmented root point clouds based on the skeleton structure.

Ground truth. By benchmarking our automated measurements against manual annotations, we validate the effectiveness of our segmentation and skeletonization methods. Ground truth values were manually extracted by a researcher proficient in both plant biology and 3D point cloud processing using [CloudCompare \(2024\)](#), a widely used tool in 3D point cloud tasks. Specifically, the Trace Polyline tool in CloudCompare was used to manually delineate the primary and lateral roots as polylines. The corresponding root lengths were then extracted directly from the Length field in the Properties panel. This manual annotation process served as a reliable reference for evaluating the accuracy of our automated pipeline.

The number of lateral roots is defined as the number of detected starting points, which serve as entry locations for the lateral root path tracking process. To evaluate lateral root detection performance, we define: *True Positives (TPs)* are starting points detected that match real lateral roots and resulted in a sufficiently complete path. *False negatives (FNs)* as true lateral roots that were either missed entirely due to undetected starting points or only partially reconstructed due to premature termination of the path. *False positives (FPs)* as detected paths that do not correspond to any real lateral root. These include paths that originate from incorrectly identified starting points or that deviate toward neighboring branches. While such deviations may resemble incomplete reconstructions, they represent incorrect path associations and are thus considered false positives.

Number of lateral roots. [Fig. 14a](#) evaluates the accuracy of lateral root detection by comparing the number of lateral roots identified by our method against ground truth counts. Our method achieved a precision of 0.90, recall of 0.86, and an F_1 score of 0.88. The fitted line ($y = 0.99x + 0.01$) closely follows the identity line ($y = x$), indicating minimal bias and strong agreement.

Lateral root angle distribution. [Fig. 14b](#) presents the angular distribution of lateral roots from five representative root systems, where

each radial line represents the branching angle between a lateral root and its associated primary root. The majority of lateral roots extend downward (angles $< 90^\circ$), consistent with biological expectations. Some upward-pointing roots are observed, likely due to deformation from root drying and extraction processes.

Primary root length. [Fig. 14c](#) validates our primary root length estimation. Our method achieved an R^2 value of 0.992, mean absolute error (MAE) of 0.50 cm, and root mean square error (RMSE) of 0.62 cm. The close alignment with the identity line and low error values demonstrate excellent consistency with manual measurements. Primary root length is a key trait in breeding for deeper rooting phenotypes and improved drought resilience.

Lateral root length. [Fig. 14d](#) shows the validation of lateral root length estimation. Five random lateral roots were selected per root system, resulting in 120 measurements. Our method achieved an R^2 of 0.987, MAE of 0.36 cm, and RMSE of 0.55 cm. The fitted line ($y = 0.97x + 0.47$) again indicates strong agreement with manual references, supporting the robustness of our length extraction.

Lateral root surface area. [Fig. 14e](#) compares the lateral root surface area computed via our tube-based reconstruction against alpha surface area estimation, which serves as a geometric cross-validation reference. The two methods exhibit high agreement with an R^2 of 0.953, MAE of 0.67 mm², and RMSE of 1.18 mm². As alpha shapes are prone to slight overestimation due to convex hull inclusion of peripheral space, the consistently lower estimates from our tube method suggest more anatomically faithful surface area calculations—further confirming the effectiveness of our tube-based reconstruction strategy.

Lateral root volume. [Fig. 14f](#) compares lateral root volumes computed using our tube-based approach versus voxel-based estimation. Again, 120 roots were evaluated. The correlation between the two methods is strong ($R^2 = 0.912$, MAE = 0.27 mm³, RMSE = 0.53 mm³). The voxel-based method slightly underestimates volume, likely due to

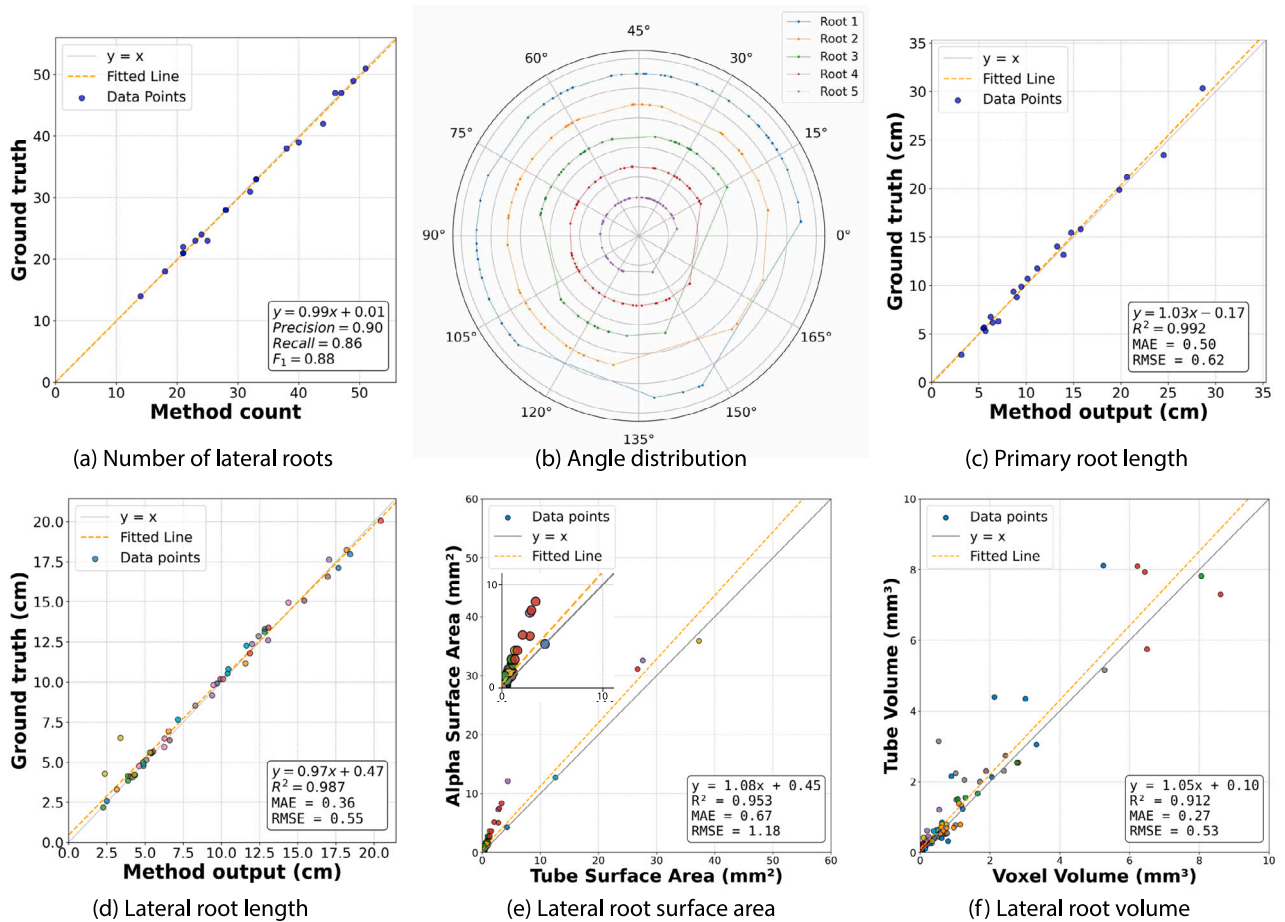


Fig. 14. Statistical analysis of extracted traits. (a) Number of lateral roots detected. (b) Angular distribution of lateral roots from five selected root systems. (c) Primary root length estimation. (d) Lateral root length estimation. (e) Comparison between tube-based and alpha shape-based surface area estimates for lateral roots. (f) Comparison between tube-based and voxel-based volume estimates for lateral roots.

small voxel resolution and the hollow center characteristic of our point cloud representation. In contrast, our tube-based approach preserves the full radius along the root's length, providing a more complete estimate of physical volume.

3.4. Limitations and future works

Despite its effectiveness in segmenting high-precision 3D root system skeletons, our root system parser remains a rule-based approach rather than a learning-based AI model, which inherently limits its generalization capacity. Our method has been validated on 30-day-old soybean root systems grown under free-field conditions, where the root system exhibits relatively predictable outward growth. While this approach could be extended to simpler tree-like structures, such as apple tree or cotton plant point clouds, with minor modifications, it is less suitable for complex architectures like maize root systems or soybean root systems grown in pots. In controlled pot conditions, roots often spiral along the container walls upon contact, violating our assumption of outward growth, which hinders accurate path extraction. Additionally, our framework is currently designed to parse only first-order lateral roots, leaving higher-order root branching unaccounted for.

Deep learning-based 3D segmentation methods, such as 3D U-Net (Çiçek et al., 2016) and Point Transformers (Wu et al., 2024), have demonstrated state-of-the-art performance on common 3D manifold objects, including chairs, couches, and televisions in indoor scenes (Armeni et al., 2019; Dai et al., 2017), as well as vehicles and pedestrians in outdoor driving datasets (Geiger et al., 2012; Caesar et al., 2020).

However, AI applications in 3D plant phenotyping remain significantly underexplored, primarily due to the lack of large-scale, annotated datasets. Our method provides an automated and scalable solution for generating labeled 3D RSA data, which could facilitate the development of deep learning models with improved generalization for end-to-end plant phenotyping. Future work could focus on leveraging our parser to construct specialized 3D root system datasets, enabling the training of AI-driven segmentation and skeletonization models that surpass rule-based methods in adaptability and robustness across diverse growth conditions.

4. Conclusion

In this work, we developed a biologically inspired root system parser that enables robust and anatomically accurate skeletonization and segmentation of 3D RSA point cloud data. By tracing natural root growth and integrating adaptive pathfinding, localized parameter tuning, and combination sorting, our method achieves precise skeleton-wise and point cloud-wise segmentation of primary and individual lateral root structures, even under conditions of noise, sparsity, and structural overlap.

The extracted skeletons and segments serve as a reliable structural backbone for quantitative trait analysis. Root length can be directly computed from the skeleton paths, while branching angles are efficiently derived by applying PCA to each skeleton segment. Building upon these skeletons, we reconstruct anatomically realistic tube-like meshes that enable analytical estimation of lateral root surface area and volume. These traits were validated against alpha shape-based surface

area calculations, voxel-based volume estimates, and manually measured root lengths. The strong agreement across all validation methods confirms both the biological fidelity and quantitative reliability of our framework.

Overall, this work advances 3D root system analysis by providing a scalable and generalizable framework for skeletonization, segmentation, and trait extraction. The ability to derive detailed, biologically meaningful traits from 3D RSA data supports broader applications in plant biology, functional genomics, and breeding. In the future, this pipeline lays the groundwork for generating large-scale annotated datasets to support the development of AI-driven models for end-to-end root system analysis across diverse species and growth environments.

CRediT authorship contribution statement

Xuehai Zhou: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Tianzi Yang:** Software, Methodology, Investigation, Formal analysis. **Rui Xu:** Software, Investigation, Data curation. **Alexander Bucksch:** Writing – review & editing, Validation. **Pierre Dutilleul:** Writing – review & editing, Supervision. **Davoud Torkamaneh:** Project administration, Funding acquisition. **Shangpeng Sun:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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